

LOGICAL STRUCTURES

Lecture 1. Declarative sentences and Natural deduction

Advanced Program in Computer Science

Spring, 2025

- ① Declarative sentences
- ② Natural deduction
 - Rules for natural deduction
 - Derived rules
 - Provable equivalence
 - Proof by contradiction

Examples

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Example 1.3

If Superman were able and willing to prevent evil, he would do so. If Superman were unable to prevent evil, he would be impotent; if he were unwilling to prevent evil, he would be malevolent. Superman does not prevent evil. If Superman exists, he is neither impotent nor malevolent. *Therefore*, Superman does not exist.

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- 3 Every even natural number > 2 is the sum of two prime numbers (Goldbach's conjecture).
- 4 All Martians like pepperoni on their pizza.

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- ❻ May fortune come your way.

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- 1 Consider certain declarative sentences as being *atomic*, or *indecomposable*, like the sentence
‘The number 5 is even.’
- 2 Assign certain distinct symbols $p, q, r, \dots$ to each of these atomic sentences and we can then code up more complex sentences in a *compositional* way.

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Consider some atomic sentences:

p : 'I won the lottery last week.'

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'If I won the lottery last week, then I purchased a lottery ticket'

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$\neg$ binds more tightly than $\vee$ and $\wedge$,

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Binding priorities

Exercise: The formulas of propositional logic below implicitly assume the binding priorities of the logical connectives put forward in Convention. Make sure that you fully understand those conventions by reinserting as many brackets as possible.

❶ $\neg p \wedge q \rightarrow r$

❷ $p \vee q \rightarrow \neg p \wedge r$

❸ $(p \rightarrow q) \rightarrow r \rightarrow s \vee t$

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- Suppose we have a set of formulas $\phi_1, \phi_2, \dots, \phi_n$, **premises**, and another formula ψ , **conclusion**.

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- This expression is called a **sequent**; it is **valid** if a proof for it can be found

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- **Fallacy** is form of incorrect reasoning which leads to invalid sequent.

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Greek letters

Lower-case

ϕ	phi
ψ	psi
χ	chi
η	eta
α	alpha
β	beta
γ	gamma

Upper-case

Φ	Phi
Ψ	Psi
Γ	Gamma
Δ	Delta

Rules for natural deduction

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- The rules for conjunction
- The rules of double negation
- The rule implies introduction
- The rules for disjunction
- The rules for negation

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- 1 $\wedge e_1$: if we have a proof for $\phi \wedge \psi$ then by apply this rule we get a proof for ϕ ;
- 2 $\wedge e_2$: if we have a proof for $\phi \wedge \psi$ then by apply this rule we get a proof for ψ .

The rules for conjunction

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Use rules for conjunction to prove that $p \wedge q, r \vdash q \wedge r$ is valid.

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Use rules for conjunction to prove that $p \wedge q, r \vdash q \wedge r$ is valid.

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$$q \wedge r$$

- fill the gap between the premises and the conclusion by applying a suitable sequence of proof rules

$$1 \qquad p \wedge q \quad \text{premise}$$

$$2 \qquad r \quad \text{premise}$$

$$3 \qquad q \quad \wedge e_2 \ 1$$

$$4 \qquad q \wedge r \quad \wedge i \ 3, 2$$

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- fill the gap between the premises and the conclusion by applying a suitable sequence of proof rules

1	$p \wedge q$	premise
2	r	premise
3	q	$\wedge e_2$ 1
4	$q \wedge r$	$\wedge i$ 3, 2

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- proof in tree structure

$$\begin{array}{c}
 \frac{p \wedge q}{q} \wedge e_2 \qquad r \\
 \hline
 q \wedge r \quad \wedge i
 \end{array}$$

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- rules of elimination and introduction for double negation:

$$\frac{\neg\neg\phi}{\phi} \neg\neg\text{e} \qquad \frac{\phi}{\neg\neg\phi} \neg\neg\text{i}$$

The rules for double negation

Example 2.4

The proof for the sequent $p, \neg\neg(q \wedge r) \vdash \neg\neg p \wedge r$.

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1	p	premise
2	$\neg\neg(q \wedge r)$	premise
3	$\neg\neg p$	$\neg\neg$ i 1
4	$q \wedge r$	$\neg\neg$ e 2
5	r	$\wedge$ e ₂ 4
6	$\neg\neg p \wedge r$	$\wedge$ i 3, 5

The rules for double negation

Example 2.5

The proof for the sequent $(p \wedge q) \wedge r, s \wedge t \vdash q \wedge s$.

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The proof for the sequent $(p \wedge q) \wedge r, s \wedge t \vdash q \wedge s$.

1	$(p \wedge q) \wedge r$	premise
2	$s \wedge t$	premise
3	$p \wedge q$	$\wedge e_1$ 1
4	q	$\wedge e_2$ 3
5	s	$\wedge e_1$ 2
6	$q \wedge s$	$\wedge i$ 4, 5

The rule for eliminating implication

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- parameters ϕ and ψ for $\rightarrow e$ can be instantiated to any sentence

1	$\neg p \wedge q$	premise
2	$\neg p \wedge q \rightarrow r \vee \neg p$	premise
3	$r \vee \neg p$	$\rightarrow e$ 2, 1

The rule for eliminating implication

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The proof for the sequent $p, p \rightarrow q, p \rightarrow (q \rightarrow r) \vdash r$

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The proof for the sequent $p, p \rightarrow q, p \rightarrow (q \rightarrow r) \vdash r$

1	$p \rightarrow (q \rightarrow r)$	premise
2	$p \rightarrow q$	premise
3	p	premise
4	$q \rightarrow r$	$\rightarrow$ e 1, 3
5	q	$\rightarrow$ e 2, 3
6	r	$\rightarrow$ e 4, 5

Modus Tollens

$$\frac{\phi \rightarrow \psi \quad \neg\psi}{\neg\phi} \text{ MT}$$

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Example 2.7

‘If Abraham Lincoln was Ethiopian, then he was African.
Abraham Lincoln was not African.

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‘If Abraham Lincoln was Ethiopian, then he was African.
Abraham Lincoln was not African.
Therefore, he was not Ethiopian.’

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Example 2.8

Proof for the sequent $p \rightarrow (q \rightarrow r), p, \neg r \vdash \neg q$

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Proof for the sequent $p \rightarrow (q \rightarrow r), p, \neg r \vdash \neg q$

1	$p \rightarrow (q \rightarrow r)$	premise
2	p	premise
3	$\neg r$	premise
4	$q \rightarrow r$	$\rightarrow$ e 1, 2
5	$\neg q$	MT 4, 3

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Example 2.9

Combination of MT with $\neg\neg e$ or $\neg\neg i$

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3	$\neg\neg q$	$\neg\neg i$ 2
4	$\neg p$	MT 1, 3

shows the validity of the sequent $p \rightarrow \neg q, q \vdash \neg p$.

Exercises

Prove the validity of the following sequents.

① $p \rightarrow (p \rightarrow q), p \vdash q.$

② $q \rightarrow (p \rightarrow r), \neg r, q \vdash \neg p$

Rule implies introduction

Rule implies introduction

- MT shows that $p \rightarrow q, \neg q \vdash \neg p$ is valid.

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- MT shows that $p \rightarrow q, \neg q \vdash \neg p$ is valid.
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1	$p \rightarrow q$	premise
2	$\neg q$	assumption
3	$\neg p$	MT 1, 2
4	$\neg q \rightarrow \neg p$	$\rightarrow$ i 2–3

Rule implies introduction

- MT shows that $p \rightarrow q, \neg q \vdash \neg p$ is valid.
- Is $p \rightarrow q \vdash \neg q \rightarrow \neg p$ valid? How to proof?
- **Mechanics:** Suppose that $p \rightarrow q$ is true. If we *temporarily* assume that $\neg q$ holds, we can use MT to infer $\neg p$. Thus assuming $p \rightarrow q$ we can show that $\neg q$ **implies** $\neg p$.

1	$p \rightarrow q$	premise
2	$\neg q$	assumption
3	$\neg p$	MT 1, 2
4	$\neg q \rightarrow \neg p$	$\rightarrow$ i 2-3

$$\frac{\begin{array}{c} \phi \\ \vdots \\ \psi \end{array}}{\phi \rightarrow \psi} \rightarrow\text{i}$$

Rule implies introduction

Example 2.10

Proof using $\rightarrow i$

Rule implies introduction

Example 2.10

Proof using $\rightarrow i$

1	$\neg q \rightarrow \neg p$	premise
2	p	assumption
3	$\neg\neg p$	$\neg\neg i$ 2
4	$\neg\neg q$	MT 1, 3
5	$p \rightarrow \neg\neg q$	$\rightarrow i$ 2–4

Rule implies introduction

Example 2.10

Proof using $\rightarrow i$

1	$\neg q \rightarrow \neg p$	premise
2	p	assumption
3	$\neg\neg p$	$\neg\neg i$ 2
4	$\neg\neg q$	MT 1, 3
5	$p \rightarrow \neg\neg q$	$\rightarrow i$ 2–4

shows the validity of the sequent $\neg q \rightarrow \neg p \vdash p \rightarrow \neg\neg q$.

Definition 2.11

Logical formulas ϕ with valid sequent $\vdash \psi$ are *theorems*.

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Logical formulas ϕ with valid sequent $\vdash \psi$ are *theorems*.

Example 2.12

Theorem whose proof utilises most of the rules introduced so far

1	$q \rightarrow r$	assumption
2	$\neg q \rightarrow \neg p$	assumption
3	p	assumption
4	$\neg \neg p$	$\neg$ -i 3
5	$\neg \neg q$	MT 2, 4
6	q	$\neg$ -e 5
7	r	$\rightarrow$ -e 1, 6
8	$p \rightarrow r$	$\rightarrow$ -i 3–7
9	$(\neg q \rightarrow \neg p) \rightarrow (p \rightarrow r)$	$\rightarrow$ -i 2–8
10	$(q \rightarrow r) \rightarrow ((\neg q \rightarrow \neg p) \rightarrow (p \rightarrow r))$	$\rightarrow$ -i 1–9

Therefore the sequent $\vdash (q \rightarrow r) \rightarrow ((\neg q \rightarrow \neg p) \rightarrow (p \rightarrow r))$ is valid,

Therefore the sequent $\vdash (q \rightarrow r) \rightarrow ((\neg q \rightarrow \neg p) \rightarrow (p \rightarrow r))$ is valid, showing that

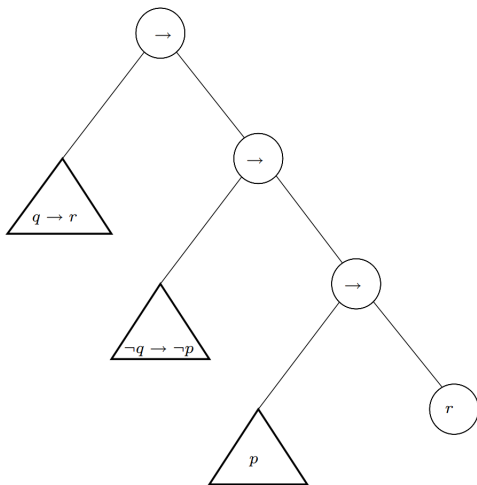
$$(q \rightarrow r) \rightarrow ((\neg q \rightarrow \neg p) \rightarrow (p \rightarrow r))$$

is a theorem.

Therefore the sequent $\vdash (q \rightarrow r) \rightarrow ((\neg q \rightarrow \neg p) \rightarrow (p \rightarrow r))$ is valid, showing that

$$(q \rightarrow r) \rightarrow ((\neg q \rightarrow \neg p) \rightarrow (p \rightarrow r))$$

is a theorem.



Example 2.13

Prove the validity of the sequent $p \wedge q \rightarrow r \vdash p \rightarrow (q \rightarrow r)$

Example 2.13

Prove the validity of the sequent $p \wedge q \rightarrow r \vdash p \rightarrow (q \rightarrow r)$

$$p \wedge q \rightarrow r \vdash p \rightarrow (q \rightarrow r):$$

1	$p \wedge q \rightarrow r$	premise
2	p	assumption
3	q	assumption
4	$p \wedge q$	$\wedge i$ 2, 3
5	r	$\rightarrow e$ 1, 4
6	$q \rightarrow r$	$\rightarrow i$ 3–5
7	$p \rightarrow (q \rightarrow r)$	$\rightarrow i$ 2–6

Example 2.14

Prove the validity of the ‘converse’ of the previous sequent

$$p \rightarrow (q \rightarrow r) \vdash p \wedge q \rightarrow r$$

Example 2.14

Prove the validity of the ‘converse’ of the previous sequent

$$p \rightarrow (q \rightarrow r) \vdash p \wedge q \rightarrow r$$

1	$p \rightarrow (q \rightarrow r)$	premise
2	$p \wedge q$	assumption
3	p	$\wedge e_1$ 2
4	q	$\wedge e_2$ 2
5	$q \rightarrow r$	$\rightarrow e$ 1, 3
6	r	$\rightarrow e$ 5, 4
7	$p \wedge q \rightarrow r$	$\rightarrow i$ 2–6

Example 2.14

Prove the validity of the ‘converse’ of the previous sequent

$$p \rightarrow (q \rightarrow r) \vdash p \wedge q \rightarrow r$$

1	$p \rightarrow (q \rightarrow r)$	premise
2	$p \wedge q$	assumption
3	p	$\wedge e_1$ 2
4	q	$\wedge e_2$ 2
5	$q \rightarrow r$	$\rightarrow e$ 1, 3
6	r	$\rightarrow e$ 5, 4
7	$p \wedge q \rightarrow r$	$\rightarrow i$ 2–6

We say that those two formulas are *equivalent*

$$p \wedge q \rightarrow r \dashv\vdash p \rightarrow (q \rightarrow r)$$

Example 2.15

Proof that uses introduction and-elimination rules for conjunction; it shows the validity of the sequent

$$p \rightarrow q \vdash p \wedge r \rightarrow q \wedge r$$

Example 2.15

Proof that uses introduction and-elimination rules for conjunction; it shows the validity of the sequent

$$p \rightarrow q \vdash p \wedge r \rightarrow q \wedge r$$

1	$p \rightarrow q$	premise
2	$p \wedge r$	assumption
3	p	$\wedge e_1$ 2
4	r	$\wedge e_2$ 2
5	q	$\rightarrow e$ 1, 3
6	$q \wedge r$	$\wedge i$ 5, 4
7	$p \wedge r \rightarrow q \wedge r$	$\rightarrow i$ 2–6

Exercises

Exercise 1. Prove the validity of the following sequents:

① $p \vdash (p \rightarrow q) \rightarrow q$

② $(p \rightarrow r) \wedge (q \rightarrow r) \vdash p \wedge q \rightarrow r$

Exercises

Exercise 1. Prove the validity of the following sequents:

- ① $p \vdash (p \rightarrow q) \rightarrow q$
- ② $(p \rightarrow r) \wedge (q \rightarrow r) \vdash p \wedge q \rightarrow r$

Exercise 2. Check the validity of the following sequents

- ① $\neg p \rightarrow \neg q \vdash q \rightarrow p$
- ② $(p \rightarrow r) \wedge (q \rightarrow r) \vdash p \wedge q \rightarrow \neg r$

Rules for Disjunction

Rules for Disjunction

- *or-introduction*: introduction of disjunction

$$\frac{\phi}{\phi \vee \psi} \vee i_1 \qquad \frac{\psi}{\phi \vee \psi} \vee i_2$$

Rules for Disjunction

- *or-elimination:*

Rules for Disjunction

- *or-elimination*: Suppose that we want to show some proposition χ by assuming $\phi \vee \psi$, we have to give two separate proofs which need to combine into one argument:

Rules for Disjunction

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 - First, we assume ϕ is true and have to come up with a proof of χ .
 - Next, we assume ψ is true and need to give a proof of χ as well.
 - Given these two proofs, we can infer χ from the truth of $\phi \vee \psi$, since our case analysis above is exhaustive.

Rules for Disjunction

- **or-elimination:** Suppose that we want to show some proposition χ by assuming $\phi \vee \psi$, we have to give two separate proofs which need to combine into one argument:
 - First, we assume ϕ is true and have to come up with a proof of χ .
 - Next, we assume ψ is true and need to give a proof of χ as well.
 - Given these two proofs, we can infer χ from the truth of $\phi \vee \psi$, since our case analysis above is exhaustive.

$$\begin{array}{c}
 \begin{array}{|c|} \hline \phi \\ \hline \vdots \\ \hline \chi \\ \hline \end{array}
 \quad
 \begin{array}{|c|} \hline \psi \\ \hline \vdots \\ \hline \chi \\ \hline \end{array} \\
 \hline
 \phi \vee \psi \qquad \qquad \qquad \chi \qquad \vee e
 \end{array}$$

Rules for Disjunction

Remark 2.1

$p \vee q \vdash q \vee p$ is valid.

Rules for Disjunction

Remark 2.1

$p \vee q \vdash q \vee p$ is valid.

1	$p \vee q$	premise
2	p	assumption
3	$q \vee p$	$\vee i_2$ 2
4	q	assumption
5	$q \vee p$	$\vee i_1$ 4
6	$q \vee p$	$\vee e$ 1, 2–3, 4–5

Examples

Example 2.16

Prove that the sequent $q \rightarrow r \vdash p \vee q \rightarrow p \vee r$ is valid

Examples

Example 2.16

Prove that the sequent $q \rightarrow r \vdash p \vee q \rightarrow p \vee r$ is valid

1	$q \rightarrow r$	premise
2	$p \vee q$	assumption
3	p	assumption
4	$p \vee r$	$\vee i_1$ 3
5	q	assumption
6	r	$\rightarrow e$ 1, 5
7	$p \vee r$	$\vee i_2$ 6
8	$p \vee r$	$\vee e$ 2, 3–4, 5–7
9	$p \vee q \rightarrow p \vee r$	$\rightarrow i$ 2–8

Examples

Example 2.17

Prove that the sequent $(p \vee q) \vee r \vdash p \vee (q \vee r)$ is valid

Examples

Example 2.17

Prove that the sequent $(p \vee q) \vee r \vdash p \vee (q \vee r)$ is valid

1	$(p \vee q) \vee r$	premise
2	$p \vee q$	assumption
3	p	assumption
4	$p \vee (q \vee r)$	$\vee i_1$ 3
5	q	assumption
6	$q \vee r$	$\vee i_1$ 5
7	$p \vee (q \vee r)$	$\vee i_2$ 6
8	$p \vee (q \vee r)$	$\vee e$ 2, 3–4, 5–7
9	r	assumption
10	$q \vee r$	$\vee i_2$ 9
11	$p \vee (q \vee r)$	$\vee i_2$ 10
12	$p \vee (q \vee r)$	$\vee e$ 1, 2–8, 9–11

Example 2.18

Prove that the sequent $p \wedge (q \vee r) \vdash (p \wedge q) \vee (p \wedge r)$ is valid.

Example 2.18

Prove that the sequent $p \wedge (q \vee r) \vdash (p \wedge q) \vee (p \wedge r)$ is valid.

1	$p \wedge (q \vee r)$	premise
2	p	$\wedge e_1$ 1
3	$q \vee r$	$\wedge e_2$ 1
4	q	assumption
5	$p \wedge q$	$\wedge i$ 2, 4
6	$(p \wedge q) \vee (p \wedge r)$	$\vee i_1$ 5
7	r	assumption
8	$p \wedge r$	$\wedge i$ 2, 7
9	$(p \wedge q) \vee (p \wedge r)$	$\vee i_2$ 8
10	$(p \wedge q) \vee (p \wedge r)$	$\vee e$ 3, 4–6, 7–9

Example 2.18

Prove that the sequent $p \wedge (q \vee r) \vdash (p \wedge q) \vee (p \wedge r)$ is valid.

1	$p \wedge (q \vee r)$	premise
2	p	$\wedge e_1$ 1
3	$q \vee r$	$\wedge e_2$ 1
4	q	assumption
5	$p \wedge q$	$\wedge i$ 2, 4
6	$(p \wedge q) \vee (p \wedge r)$	$\vee i_1$ 5
7	r	assumption
8	$p \wedge r$	$\wedge i$ 2, 7
9	$(p \wedge q) \vee (p \wedge r)$	$\vee i_2$ 8
10	$(p \wedge q) \vee (p \wedge r)$	$\vee e$ 3, 4–6, 7–9

Exercise

Prove that the sequent $(p \wedge q) \vee (p \wedge r) \vdash p \wedge (q \vee r)$ is valid.

Example 2.18

Prove that the sequent $p \wedge (q \vee r) \vdash (p \wedge q) \vee (p \wedge r)$ is valid.

1	$p \wedge (q \vee r)$	premise
2	p	$\wedge e_1$ 1
3	$q \vee r$	$\wedge e_2$ 1
4	q	assumption
5	$p \wedge q$	$\wedge i$ 2, 4
6	$(p \wedge q) \vee (p \wedge r)$	$\vee i_1$ 5
7	r	assumption
8	$p \wedge r$	$\wedge i$ 2, 7
9	$(p \wedge q) \vee (p \wedge r)$	$\vee i_2$ 8
10	$(p \wedge q) \vee (p \wedge r)$	$\vee e$ 3, 4–6, 7–9

Exercise

Prove that the sequent $(p \wedge q) \vee (p \wedge r) \vdash p \wedge (q \vee r)$ is valid.

Rule for copy

Rule for copy

Example 2.19

Prove that the sequent $\vdash p \rightarrow (q \rightarrow p)$ is valid.

Rule for copy

Example 2.19

Prove that the sequent $\vdash p \rightarrow (q \rightarrow p)$ is valid.

1	p	assumption
2	q	assumption
3	p	copy 1
4	$q \rightarrow p$	$\rightarrow i$ 2–3
5	$p \rightarrow (q \rightarrow p)$	$\rightarrow i$ 1–4

Rules for Negation

Rules for Negation

Definition 2.20

Contradictions are expressions of the form $\phi \wedge \neg\phi$ or $\neg\phi \wedge \phi$, where ϕ is any formula.

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- **Note:** Any formula can be derived from a contradiction

$$p \wedge \neg p \vdash q$$

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- Rule of **bottom-elimination** (contradiction):

$$\frac{\perp}{\phi} \perp e$$

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Contradictions are expressions of the form $\phi \wedge \neg\phi$ or $\neg\phi \wedge \phi$, where ϕ is any formula.

- **Note:** Any formula can be derived from a contradiction

$$p \wedge \neg p \vdash q$$

- Rule of **bottom-elimination** (contradiction):

$$\frac{\perp}{\phi} \perp e$$

- **Negation-elimination:**

$$\frac{\phi \quad \neg\phi}{\perp} \neg e$$

Examples

Example 2.21

Show that $\neg p \vee q \vdash p \rightarrow q$ is valid.

Examples

Example 2.21

Show that $\neg p \vee q \vdash p \rightarrow q$ is valid.

1	$\neg p \vee q$		
2	$\neg p$	assumption	q assumption
3	p	assumption	p assumption
4	$\perp$	$\neg$ e 3, 2	q copy 2
5	q	$\perp$ e 4	
6	$p \rightarrow q$	$\rightarrow$ i 3–5	$p \rightarrow q$ $\rightarrow$ i 3–4
7	$p \rightarrow q$		$\vee$ e 1, 2–6

Negation-introduction

- *Negation-introduction:*

$$\frac{\boxed{\begin{array}{c} \phi \\ \vdots \\ \bot \end{array}}}{\neg\phi} \neg\text{i.}$$

Examples

Example 2.22

Show that $p \rightarrow q, p \rightarrow \neg q \vdash \neg p$ is valid.

Examples

Example 2.22

Show that $p \rightarrow q, p \rightarrow \neg q \vdash \neg p$ is valid.

1	$p \rightarrow q$	premise
2	$p \rightarrow \neg q$	premise
3	p	assumption
4	q	$\rightarrow$ e 1, 3
5	$\neg q$	$\rightarrow$ e 2, 3
6	$\perp$	$\neg$ e 4, 5
7	$\neg p$	$\neg$ i 3–6

Examples

Example 2.23

Show that $p \rightarrow (q \rightarrow r), p, \neg r \vdash \neg q$ is valid.

Examples

Example 2.23

Show that $p \rightarrow (q \rightarrow r), p, \neg r \vdash \neg q$ is valid.

1	$p \rightarrow (q \rightarrow r)$	premise
2	p	premise
3	$\neg r$	premise
4	q	assumption
5	$q \rightarrow r$	$\rightarrow$ e 1, 2
6	r	$\rightarrow$ e 5, 4
7	$\perp$	$\neg$ e 6, 3
8	$\neg q$	$\neg$ i 4–7

Examples

Example 2.24

Show that $p \wedge \neg q \rightarrow r, \neg r, p \vdash q$ is valid.

Examples

Example 2.24

Show that $p \wedge \neg q \rightarrow r, \neg r, p \vdash q$ is valid.

1	$p \wedge \neg q \rightarrow r$	premise
2	$\neg r$	premise
3	p	premise
4	$\neg q$	assumption
5	$p \wedge \neg q$	$\wedge$ i 3, 4
6	r	$\rightarrow$ e 1, 5
7	$\perp$	$\neg$ e 6, 2
8	$\neg\neg q$	$\neg$ i 4–7
9	q	$\neg\neg$ e 8

Derived Rules

$$\frac{\phi \rightarrow \psi \quad \neg\psi}{\neg\phi} \text{ MT}$$

Modus Tollens

is *not a primitive* rule of natural deduction, but it can be *derived* from some of the other rules.

Derived Rules

$$\frac{\phi \rightarrow \psi \quad \neg\psi}{\neg\phi} \text{ MT}$$

Modus Tollens is *not a primitive* rule of natural deduction, but it can be *derived* from some of the other rules. For example, using $\rightarrow e$, $\neg e$ and $\neg i$ as below

1	$\phi \rightarrow \psi$	premise
2	$\neg\psi$	premise
3	ϕ	assumption
4	ψ	$\rightarrow e$ 1, 3
5	$\perp$	$\neg e$ 4, 2
6	$\neg\phi$	$\neg i$ 3–5

Derived Rules

Two important derived rules are:

- 1 *Proof by contradiction:*

Derived Rules

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- 1 ***Proof by contradiction***: (PBC for short, Latin name *reductio ad absurdum*) if from $\neg\phi$ we obtain a contradiction, then we are entitled to deduce ϕ :

Derived Rules

Two important derived rules are:

- 1 **Proof by contradiction:** (PBC for short, Latin name *reductio ad absurdum*) if from $\neg\phi$ we obtain a contradiction, then we are entitled to deduce ϕ :

$$\frac{\begin{array}{|c|} \hline \neg\phi \\ \vdots \\ \perp \\ \hline \end{array}}{\phi} \text{ PBC.}$$

Derived Rules

PBC can be derived from $\rightarrow i$, $\neg i$, $\rightarrow e$ and $\neg\neg e$ as follows

1	$\neg\phi \rightarrow \perp$	given
2	$\neg\phi$	assumption
3	$\perp$	$\rightarrow e$ 1, 2
4	$\neg\neg\phi$	$\neg i$ 2–3
5	ϕ	$\neg\neg e$ 4

② *Law of Excluded Middle:*

- ② **Law of Excluded Middle:** (LEM for short, Latin name *tertium non datur* or simply *no third*) $\phi \vee \neg\phi$ is true.

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Example: if-statements in programming language

‘if B $\{C_1\}$ else $\{C_2\}$ ’

- ② **Law of Excluded Middle:** (LEM for short, Latin name *tertium non datur* or simply *no third*) $\phi \vee \neg\phi$ is true.

Example: if-statements in programming language

‘if B $\{C_1\}$ else $\{C_2\}$ ’

1	$\neg(\phi \vee \neg\phi)$	assumption
2	ϕ	assumption
3	$\phi \vee \neg\phi$	$\vee i_1$ 2
4	$\perp$	$\neg e$ 3, 1
5	$\neg\phi$	$\neg i$ 2–4
6	$\phi \vee \neg\phi$	$\vee i_2$ 5
7	$\perp$	$\neg e$ 6, 1
8	$\neg\neg(\phi \vee \neg\phi)$	$\neg i$ 1–7
9	$\phi \vee \neg\phi$	$\neg\neg e$ 8

Example 2.25

Using LEM, we show that $p \rightarrow q \vdash \neg p \vee q$ is valid.

Example 2.25

Using LEM, we show that $p \rightarrow q \vdash \neg p \vee q$ is valid.

1	$p \rightarrow q$	premise
2	$\neg p \vee p$	LEM
3	$\neg p$	assumption
4	$\neg p \vee q$	$\vee i_1$ 3
5	p	assumption
6	q	$\rightarrow e$ 1, 5
7	$\neg p \vee q$	$\vee i_2$ 6
8	$\neg p \vee q$	$\vee e$ 2, 3–4, 5–7

Example 2.25

Using LEM, we show that $p \rightarrow q \vdash \neg p \vee q$ is valid.

1	$p \rightarrow q$	premise
2	$\neg p \vee p$	LEM
3	$\neg p$	assumption
4	$\neg p \vee q$	$\vee i_1$ 3
5	p	assumption
6	q	$\rightarrow e$ 1, 5
7	$\neg p \vee q$	$\vee i_2$ 6
8	$\neg p \vee q$	$\vee e$ 2, 3–4, 5–7

Exercise

Show that $\vdash \neg q \vee (p \rightarrow q)$ is valid.

Example 2.25

Using LEM, we show that $p \rightarrow q \vdash \neg p \vee q$ is valid.

1	$p \rightarrow q$	premise
2	$\neg p \vee p$	LEM
3	$\neg p$	assumption
4	$\neg p \vee q$	$\vee i_1$ 3
5	p	assumption
6	q	$\rightarrow e$ 1, 5
7	$\neg p \vee q$	$\vee i_2$ 6
8	$\neg p \vee q$	$\vee e$ 2, 3–4, 5–7

Exercise

Show that $\vdash \neg q \vee (p \rightarrow q)$ is valid.

Exercise

Show that $\vdash \neg q \vee (p \rightarrow q)$ with PBC instead of LEM.

Natural Deduction: Summary

	<i>introduction</i>	<i>elimination</i>
$\wedge$	$\frac{\phi \quad \psi}{\phi \wedge \psi} \wedge i$	$\frac{\phi \wedge \psi}{\phi} \wedge e_1 \quad \frac{\phi \wedge \psi}{\psi} \wedge e_2$
$\vee$	$\frac{\phi}{\phi \vee \psi} \vee i_1 \quad \frac{\psi}{\phi \vee \psi} \vee i_2$	$\frac{\phi \vee \psi \quad \boxed{\begin{array}{c} \phi \\ \vdots \\ \chi \end{array}} \quad \boxed{\begin{array}{c} \psi \\ \vdots \\ \chi \end{array}}}{\chi} \vee e$

	<i>introduction</i>	<i>elimination</i>
$\rightarrow$	$\frac{\boxed{\begin{array}{c} \phi \\ \vdots \\ \psi \end{array}}}{\phi \rightarrow \psi} \rightarrow i$	$\frac{\phi \quad \phi \rightarrow \psi}{\psi} \rightarrow e$
$\neg$	$\frac{\boxed{\begin{array}{c} \phi \\ \vdots \\ \perp \end{array}}}{\neg \phi} \neg i$	$\frac{\phi \quad \neg \phi}{\perp} \neg e$
$\perp$	(no introduction rule for $\perp$)	$\frac{\perp}{\phi} \perp e$
$\neg\neg$		$\frac{\neg\neg\phi}{\phi} \neg\neg e$

Some useful derived rules:

$$\frac{\phi \rightarrow \psi \quad \neg\psi}{\neg\phi} \text{ MT}$$

$$\frac{\phi}{\neg\neg\phi} \neg\text{-i}$$

$$\frac{\boxed{\begin{array}{c} \neg\phi \\ \vdots \\ \perp \end{array}}}{\phi} \text{ PBC}$$

$$\frac{}{\phi \vee \neg\phi} \text{ LEM}$$

Exercise

Show that the following sequents are valid

$$\text{a) } p \vee q, \neg p \vdash q$$

$$\text{b) } p \rightarrow (q \rightarrow r) \vdash q \rightarrow (p \rightarrow r)$$

$$\text{c) } (p \rightarrow q) \rightarrow r \vdash p \rightarrow (q \rightarrow r)$$

$$\text{d) } p \vee (q \wedge r) \vdash p \vee q$$

$$\text{e) } a \vee b, a \rightarrow c, \neg d \rightarrow \neg b \vdash c \vee d$$

$$\text{f) } (a \rightarrow b) \wedge (b \rightarrow a) \vdash (a \wedge b) \vee (\neg a \wedge \neg b)$$

$$\text{g) } \vdash ((p \rightarrow q) \rightarrow p) \rightarrow p$$

Provable Equivalence

Definition 2.26

Let ϕ and ψ be formulas of propositional logic. We say that ϕ and ψ are *provably equivalent* iff the sequents $\phi \vdash \psi$ and $\psi \vdash \phi$ are valid and we denote by $\phi \dashv\vdash \psi$.

Provable Equivalence

Definition 2.26

Let ϕ and ψ be formulas of propositional logic. We say that ϕ and ψ are **provably equivalent** iff the sequents $\phi \vdash \psi$ and $\psi \vdash \phi$ are valid and we denote by $\phi \dashv\vdash \psi$.

Example 2.27

Examples of provably equivalent formulas are

$$\neg(p \wedge q) \dashv\vdash \neg q \vee \neg p$$

$$\neg(p \vee q) \dashv\vdash \neg p \wedge \neg q$$

$$p \rightarrow q \dashv\vdash \neg q \rightarrow \neg p$$

$$p \rightarrow q \dashv\vdash \neg p \vee q$$

$$p \wedge q \rightarrow r \dashv\vdash p \rightarrow (q \rightarrow r)$$

$$p \wedge q \rightarrow r \dashv\vdash p \rightarrow (q \rightarrow r)$$

Extra Reading

Section 1.2.5. Proof by contradiction in the textbook.